

Financial Fake News Detection via Context-Aware Embedding and Sequential Representation using Cross-Joint Networks

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Abstract—Fake news is a pervasive phenomenon over online social media platforms. The computation detection of fake news is a well-known problem. Financial fake news has immense potential to mislead the financial domain, but it has drawn less attention from researchers. In this paper, we introduce a new approach for financial fake news detection. Starting with preparing financial fake news datasets using topic modelling and Transformer-based techniques. We introduce two cross-joint networks. The first cross-joint network – CAEN is used to give context-aware linguistic and financial embeddings. The combined embedding outcome is passed to the second cross-joint network – CSRN which is employed to produce contextual sequential representation to detect financial fake news. We evaluate proposed approach on two benchmark datasets and it shows impressive results. We have compared the evaluation results of the proposed approach with relevant state-of-the-art and numerous neural network-based baseline methods, and it is noticed that it also shows remarkably better.

Index Terms—fake news, financial fake news, online social media, information retrieval, misleading information

I. INTRODUCTION

Online social media (OSM) platforms have been now an enriching source of information [1]. Users frequently exchange information, establish connections, and post on any ongoing events and trends across these OSM platforms, such as Twitter, Facebook, and Instagram [2]. Consequently, this generates a huge amount of user-generated data. It also includes several misleading pieces of information like fake news, satire, hoax, rumor, etc [3]. In the last few years, the rapid proliferation of fake news is designed to mislead using OSM platforms. It is a possible threats to freedom of expression. It reduces the impact of real news, induces distrust in the media, and affects sectors like finance.

A. Financial Fake News

Financial news plays an important role in finance, such as investors in taking financial and investment-related decisions, real-time financial market coverage, and stock market trading. In the last few years, fraud charges against individuals and companies are seen for spreading fake financial news. Besides that, the availability of data across OSM generated by users is a good source of financial information which can

disseminate false information. Further, fraudsters can easily use and manipulate such finance-based false and misleading information to perform unethical market actions¹. Figure 1 shows an exemplar financial fake news (FFN) related to the Google Pay operation of payment in India². It may lead to distort stock trading volumes and returns, adversely affect the social well-being of OSM platforms, and fair market competition. Therefore, FFN detection is the need of the hour and a worth research investigation problem.

Google Pay not a payment system operator: RBI to Delhi HC

The Reserve Bank of India (RBI) has told the Delhi High Court that Google Pay is a third party app provider (TPAP) and does not operate any payment systems.

Fig. 1. An exemplar financial fake news

B. Our Contributions

In most of the existing studies, the role of context is found important in fake news detection problem. Thus, manual verification like fact-checker is used to verify the news headlines content to establish ground-truth in terms of social context [3]. However, linguistics and domain-specific related context are less covered. They are important for fake news detection. Considering this, in this study, we have a new approach to detect FFN. Initially, we apply a hybrid approach by combining topic modeling-based unsupervised and DistilBERT-based [4] supervised approach to prepare FFN dataset from fake news category of two benchmark datasets (WELFake [5] and Fake News (Kaggle)³). We introduce two cross-joint networks (context-aware embedding network (CAEN) and context-sensitive representation network (CSRN)). The CAEN is consisted of two input layers (input news headline and input news content) which is crossed-joint connected with both BERT and FinText⁴

¹<https://bit.ly/3Ps9aDR>

²<https://bit.ly/3FRaxJc>

³<https://www.kaggle.com/competitions/fake-news/data>

⁴<https://www.rahimikia.com/fintext>

embeddings. Consequently, it gives linguistic-contextual information and financial-contextual information from BERT and FinText, respectively from input sources. Further, it is combined together to produce context-aware joint embedding representation and that is forwarded to CSRN network which is consisted of a convolutional neural network (CNN) which is crossed-joint with bi-directional long short-term memory (BiLSTM) followed by two attention layers. The CNN receives two embeddings joint outcome and gives semantic and syntactic financial and fake information and passed to forward and backward directions of BiLSTM to give FFN-based contextual information-based forward sequence and backward sequence, respectively. Two attention layers – forward attention and backward attention are used which produce forward and backward contextual representation based on variable-length sequences for forward sequences and backward sequences of BiLSTM, respectively. Thereafter, the joint context representation of two attention layers is passed to dense layer followed by *sigmoid* activation function which is responsible to perform classification of the input news information as either FFN or non-financial fake news (NFFN) (i.e., real/normal news).

Overall, the major contributions of this study are as follows:

- Introduced a new approach for financial fake news detection problem, wherein two cross-joint networks are proposed.
- Implemented a new technique to prepare FFN datasets.
- Employed context-aware embedding (financial and linguistics) and sequential representation using cross-joint networks for FFN detection.
- Conducted an experimental evaluation of the proposed FFN detection approach and compared with relevant work and baselines methods to examine its efficacy.

The rest of the paper is organized as follows. Section II highlights the existing works. Section III presents the proposed FFN detection approach. Section IV explains experimental setup and evaluation results. Finally, Section V concludes the paper and highlights future work.

II. RELATED WORKS

This section highlights the existing studies for fake news detection. The authors in [6] considered support vector machine (SVM) to determine real news or fake news. The authors in [7] evaluated 141 textual features for fake news detection. The author in [8] proposed a hybrid approach using SVM and naïve Bayes to predict and classify fake news via news content. The authors in [9] proposed S-HAN and stacked bidirectional Gated Recurrent Units (BiGRU) to recognize fake news. The authors in [10] proposed multi-domain fake news detection. The authors in [11] developed LSTM, CNN, and BERT for fake news identification on textual data. The authors in [12] considered phony news on Facebook by analysing user profile characteristics. The authors in [13] demonstrated CNN to automatically detecting fake news. Blockchain-based techniques are applied for fake news detection in [14]. Likewise, blockchain is used in [15]. The

authors in [16] distinguished fake news from real using textual property. Recently, The authors in [17] proposed a multi-fact CNN-LSTM model for FFN detection.

It can be noticed from the aforementioned discussion that researchers have proposed various techniques to detect fake news, but the problem of FFN detection is less addressed. However, it is an important and worth research investigation problem.

III. PROPOSED APPROACH

This section presents the brief details about the proposed FFN detection approach. Figure 2 presents the work-flow of the proposed FFN detection approach. It is considered as a binary classification problem. It implies that whether a piece of news headline and content is FFN or NFFN.

A. Data Collection and Pre-processing

We have discussed two benchmark datasets in sub-section I-B, we have downloaded both datasets, wherein each news is labelled as either fake news or real news. Also, they contain several unwanted information. Thus, we perform several pre-processing steps to clean the downloaded datasets to receive good results. In this line, we remove symbols, non-ASCII, digits, dots, extra white spaces, hexa-characters, slash, exclamation marks, quotes, punctuation marks, URLs, ampersands, and convert news into the lower-case form.

B. Data Preparation

This section presents the data preparation to obtain FFN from fake news on both datasets. It consists of two-steps process. To this end, in step-1, we apply unsupervised topic modelling technique. We consider financial keywords extraction process using Latent Dirichlet allocation (LDA) [18], an unsupervised topic modeling method to retrieve topics from the textual portion of the document. It discovers financial topics from fake news category and results in a language model which indicate how a word in a fake news headline and news content is related to each financial topics. Hence, it forms two clusters from the fake news category, wherein one cluster refers to the financial topics/keywords containing fake news, whereas another one refers to the normal fake news which is not related to financial topic/domain.

Further, in step-2, we apply supervised technique-based DistilBERT, a smaller, faster, and lighter model for enrichment of the data preparation and received fine-grained FFN dataset from financial topics/keywords containing fake news cluster. DistilBERT is compressed through knowledge distillation method. We train it on large AG corpus of news articles⁵ consisting of four classes/categories (world, sci/tech, sports, and business) to further predict FFN from business category and other fake news from (world, sci/tech, sports) categories. Finally, candidate FFN is considered for further processing in the proposed approach accordingly.

⁵http://groups.di.unipi.it/~gulli/AG_corpus_of_news_articles.html

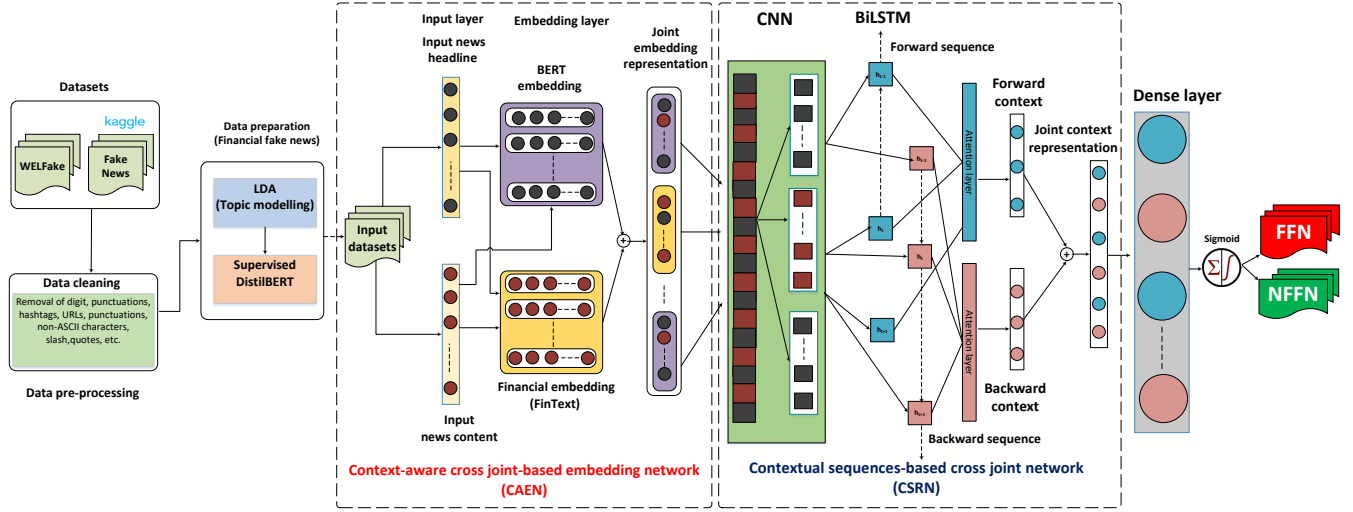


Fig. 2. Work-flow of the proposed approach for FFN detection

C. CAEN

The CAEN is crossed-joint connected with two inputs (news headline and news content) and two embeddings (BERT and FinText) layers. This network is briefly discussed in the following sub-sections:

1) *Input Layer*: Given the candidate FFN or NFFN as input headline n_h and input news content n_c consisting of w_h and w_c words, respectively. Each word from input news headline and input news content are tokenized and indexed in a dictionary. Consequently, two numeric vectors v_h and v_c are generated with varying input length size. Hence, a fixed-length h_l and c_l are maintained for input news headline vector and input news content vector, respectively. Thus, v_h and v_c are transformed to a padded-vector p_h and p_c , respectively of a fixed-length size, and then both are forwarded to BERT and FinText embeddings.

2) *Embedding Layer*: In this paper, BERT and FinText are used to learn linguistics and financial context-based word representations, respectively as low-dimensional dense vectors from two embedding spaces for both input news headline and input news content. In this line, two embeddings are cross-joint to the padded-vectors p_h and p_c , respectively. Thereafter, the resultant dense vectors are combined to give joint embedding representation.

D. CSRN

The CSRN is composed of CNN, BiLSTM, and attention layers. It is also crossed-joint network with an aim to give context-based sequential representation. This network is motivated from the [19] work. It is briefly discussed in following sub-sections:

1) *CNN Layer*: CNN performs numerous convolution operations on the input news (headline and contents). It generates global feature vector accordingly. We have used

max pool size of 3, ReLU activation function, and 64 filters with window size 3 to perform the convolutional operation to extract the financial and fake news-based semantic and syntactic feature sequence as $f_t = [f_1, f_2, \dots, f_{64}]$ from the input news and headline. Thus, equation 1 gives the n^{th} feature sequence. Finally, the filter outputs are concatenated to produce fake news and finance-based feature representation, which are passed to connect with forward LSTM and backward LSTM layers separately.

$$f_t = r(w_t \cdot x_t + b) \quad (1)$$

2) *BiLSTM Layer*: It is operational as forward LSTM in forward direction and backward LSTM in backward direction. In this paper, the forward LSTM and backward LSTM is used to generate forward latent contextual feature sequences (f_1 to f_{64}) and backward latent contextual feature sequences (f_{64} to f_1), respectively related to fake news and finance from the extracted semantic and syntactic features of the CNN layer. Equations 2 and 3 represent forward and backward directions of BiLSTM and which are passed to two attention layers in parallel.

$$\vec{l}_f = \overrightarrow{LSTM}(L_{f_t}), n \in [1, 64] \quad (2)$$

$$\overleftarrow{l}_b = \overleftarrow{LSTM}(L_{f_t}), n \in [64, 1] \quad (3)$$

3) *Attention Layer*: In this paper, two attention layers are used to highlight crucial FFN-based tokens. It emphasises on financial fake news related keyword-based features in the input news headline and content. $\vec{l}_f$ refers to word annotation and it is given to receive a hidden representation $\vec{\alpha}_f$, as given in equation 4, wherein w , b , and $\tanh$ refer to weight, bias, and hyperbolic tangent function, respectively, and it is done via one layer perceptron.

$$\vec{\alpha}_f = \tanh(w\vec{l}_f + b) \quad (4)$$

The similarity between $\vec{\alpha}_f$ and $\vec{\beta}_f$ is calculated to highlight the importance of each token in the input, where $\vec{\beta}_f$ is word-level context vector. It shows a high-level representation of the FFN token from the input news headline and content. Equation 5 represents a normalized weight $\vec{z}_f$ via *softmax*.

$$\vec{z}_f = \frac{\exp(\vec{\alpha}_f * \vec{\beta}_f)}{\sum_{i=1}^N (\exp(\vec{\alpha}_f * \vec{\beta}_f))} \quad (5)$$

The forward context representation F_c , which is computed through $\vec{l}_f$ and $\vec{z}_f$, as given in equation 6. Likewise, the backward context representation B_c is used for the backward direction, as given in equation 7 shows. Finally, a joint FFN context-based sequential representation $A_c = [F_c, B_c]$ is received via joining F_c and B_c for a given feature sequence.

$$F_c = \sum (\vec{z}_f * \vec{l}_f) \quad (6)$$

$$B_c = \sum (\vec{z}_b * \vec{l}_b) \quad (7)$$

E. Dense and Output Layers

The fully connected dense layer produces FFN-based features set from the outcome of the CSRN and divisible into two classes. The non-linear activation – *sigmoid* is applied on the top of the dense layer, and loss function – *binary cross-entropy* interprets input headline and content labelled as FFN or NFFN.

IV. EXPERIMENTAL SETUP AND RESULTS

This section presents the experimental setup details and evaluation results of the proposed FFN detection approach.

A. Datasets

We have downloaded 37106 and 35028 fake and real news, respectively from WELFake [5] dataset, and 10370 and 10350 fake and real news, respectively from Fake News (Kaggle) dataset. Table I presents the final statistics of the datasets which gives retained instances after filtering FFN. Both datasets are divided in three sets – training, validation, and testing in 70:10:20 ratio, respectively.

TABLE I
FINAL STATISTICS OF THE RETRAINED INSTANCES AFTER FILTERING FFN

Datasets	FFN	NFFN	Total
WELFake [5]	5748	5743	11491
Fake News (kaggle)	1254	1254	2508

B. Experimental and Hyperparameter Settings

This section presents the details of experimental and hyperparameter settings which are used for the empirical evaluation of the proposed FFN detection approach. The details are available in table II.

TABLE II
EXPERIMENTAL AND HYPERPARAMETER SETTINGS

Experimental settings		Hyperparameter settings	
Environment	Configuration	Parameters	Value
Operating system	Ubuntu-20.04	Embedding dimension (BERT)	768
Processor	Intel Haswell	Embedding dimension (FinText)	300
Memory	64 GB	LSTM – #neurons	128
Scripting language	Python 3.9	Dropout	0.5
Library (neural network)	Keras 2.2.4	Padding sequences	200
GPU	NVIDIA Tesla	CNN – #filters	3
Optimizer	Adam	Epoch (early stopping)	30

C. Evaluation Metrics

The proposed FFN detection approach is evaluated using commonly used metrics, such as *Precision*, *Recall*, *F-score*, and *Accuracy*, as given in equations 8 to 11, wherein terms like *True Positive* (TP) gives the count of correctly identified FFN instances. *False Positive* (FP) gives the count of incorrectly identified FFN instances. *True Negative* (TN) gives the count of correctly identified NFFN instances. Finally, *False Negative* (FN) gives the count of incorrectly identified NFFN instances.

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

$$F-score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (10)$$

$$Accuracy = \frac{TP + TN}{\#News} \quad (11)$$

D. Experimental Results and Comparative Analysis

This section presents the experimental results of the proposed approach. We have compared our approach with Zhi et al. [17] work. They considered FFN detection using CNN and LSTM. We have re-implemented their work to compare with our proposed approach. The proposed FFN detection approach is also compared with several neural network baseline methods. Table III shows that the proposed FFN detection approach outperforms as compared to Zhi et al. [17] work and baseline methods on both datasets. It shows that the consideration of cross-joint networks based on context-aware linguistics and financial embeddings and sequential representation significantly improves the classification performance of FFN detection. Figure 3 presents the visualization of the training accuracy versus validation accuracy on both datasets. It shows that the proposed approach also receives impressive results on both datasets as compared to the existing work and baseline results.

TABLE III
PERFORMANCE EVALUATION RESULTS ON WELFAKE AND FAKE NEWS (KAGGLE) DATASETS

Datasets →	WELFake [5]			Fake News (kaggle)		
Methods ↓	Precision	Recall	F-score	Precision	Recall	F-score
Proposed approach	0.94	0.95	0.94	0.85	0.90	0.87
Zhi et al. [17]	0.92	0.88	0.90	0.76	0.88	0.82
CNN	0.91	0.87	0.89	0.80	0.75	0.77
BiGRU	0.90	0.91	0.90	0.68	0.75	0.71
BiLSTM	0.90	0.89	0.89	0.67	0.71	0.69
CNN+BiGRU	0.89	0.87	0.88	0.81	0.70	0.75
CNN+BiLSTM	0.87	0.86	0.86	0.80	0.69	0.74



Fig. 3. Training accuracy versus validation accuracy on WELFake and Fake News (kaggle) datasets

V. CONCLUSION AND FUTURE WORK

This paper has presented FFN detection problem. We have prepared FFN dataset and proposed two context-aware and cross-joint networks (CAEN and CSRN) for FFN detection. The proposed approach has evaluated on two benchmark datasets and it has received better results in comparison to the relevant work and baseline methods. The evaluation of the proposed approach on multi-modal datasets could be an interesting research direction. The creation of new annotated FFN dataset is also important for future research works. The proposed approach can also be extended over multi-lingual datasets.

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